Contact

+32456825684 (Home)
benefron@gmail.com

www.linkedin.com/in/ben-efron-b19256b2 (LinkedIn)

Top Skills

Systems Neuroscience

Neuroengineering

Neuromorphic Engineering

Languages

Hebrew (Native or Bilingual)

English (Full Professional)

Certifications

edX Honor Code Certificate for Introduction to Computer Science and Programming Using Python

Honors-Awards

Best poster award

Magna cum laude

Schulich Leader Scholarships - for excellent students in STEM areas

WBI Postdoctoral Excellence Scholarship

Ben Efron

Postdoctoral Researcher | Systems Neuroscience • Neuroengineering • Neuromorphic Computing • Data Analysis | Bridging Brain Science and Technology

Leuven, Flemish Region, Belgium

Summary

I am a Postdoctoral Researcher at the University of Liège’s Brain-Inspired Computing Lab, with a PhD in Systems Neuroscience and over seven years of hands-on experience designing, building, and running complex experimental systems to study neural computation and behavior.

My work focuses on understanding and modeling how biological systems encode and process information, bridging experimental neuroscience with emerging technologies in neuroengineering and neuromorphic computing.

I have broad experience managing the entire scientific and technical pipeline — from concept and experimental design, through automation and data acquisition, to advanced analysis and communication of results.
I have built and maintained multiple experimental setups integrating neural recordings, sensors, video tracking, actuation, and real-time control, including custom hardware (PCB design, 3D printing, motor control) and software for acquisition, synchronization, and analysis.

In data analysis, I apply machine learning and advanced statistical methods (e.g., classifiers, permutation and bootstrap tests, Bayesian models, logistic regression) to uncover structure in neural and behavioral data and test hypotheses about information coding.

I am particularly drawn to systems-level questions — treating neurons, networks, and sensory systems as adaptive, information-processing entities — and to applying this perspective to technology development in neuroAI, BCI/BMI, and embedded intelligence.

I enjoy working in interdisciplinary environments, collaborating with engineers, computer scientists, and biologists to translate scientific understanding into new computational and hardware approaches.

Core strengths:
• Systems neuroscience & neural data science
• Experimental design, automation & hardware–software integration
• Large-scale data acquisition and synchronization (HD-MEA, behavior, sensors)
• Signal processing, statistical modeling & machine learning for neural data
• Translating biological insight into technology concepts
• Scientific communication & cross-disciplinary collaboration

Experience

University of Liège
Postdoctoral Researcher - Brain-Inspired Computing Lab
July 2024 - Present (1 year 5 months)
Liège, Walloon Region, Belgium

Postdoctoral researcher working at the interface of systems neuroscience, neuroengineering, and neuromorphic computing. My work bridges experimental neuroscience with software development and data-driven analysis to connect biological insights with emerging computational and hardware technologies.

Main activities and responsibilities:
• Lead experimental work in the biological branch of the Brain-Inspired Computing Lab, focusing on neuronal population activity, information processing, and real-time control.
• Develop and adapt software for experiment management, data acquisition, and analysis using imec's high-density microelectrode array (HD-MEA) systems. Integrate these tools with standard neuroscience environments such as Open Ephys to enable closed-loop control and synchronized data acquisition.
• Work closely with the imec engineering team to implement improvements for neuroscience-focused experiments, including real-time feedback, data synchronization, and workflow optimization.

• Collaborate in the NEMODAI project — a partnership between imec, VUB Brubotics, and the University of Liège — connecting neuronal recordings from cell cultures with neuromorphic and robotic systems. Act as the primary communication link between imec and ULiège.
• Apply advanced data analysis and statistical modeling (machine learning classifiers, Bayesian inference, bootstrapping, permutation tests) to interpret large-scale neural and imaging datasets.
• Provide biological and systems-level insight for a collaborative modeling project on short-term synaptic plasticity (LTP/LTD).
• Lead a computational project developing a biologically inspired framework for sensory processing and information abstraction, designed with future neuromorphic hardware implementation in mind.

Weizmann Institute of Science

7 years 3 months

PHD Researcher - Systems neuroscience

April 2017 - June 2024 (7 years 3 months)

Rehovot Area, Israel

Doctoral research in systems neuroscience focused on auditory information processing in behaving mice, combining electrophysiology, behavior, and computational analysis to study sensory coding and neural dynamics.

Main activities and responsibilities:
• Investigated auditory information processing during natural behaviors using multi-electrode electrophysiology and behavioral paradigms.
• Designed and built custom experimental setups integrating neural, sensory, and behavioral recordings with real-time control and synchronization.
• Applied signal processing, statistical modeling, and machine learning methods to decode neuronal and behavioral patterns and test information coding hypotheses.
• Taught as a Teaching Assistant in Synaptic Physiology, covering membrane models, conductance-based models (Hodgkin–Huxley), cable theory, and synaptic communication.
• Published and presented research at international conferences; received multiple awards for scientific excellence.

MSc Student

April 2017 - March 2020 (3 years)

Reẖovot, Central, Israel

Prof. Hermona Soreq laboratory

Undergraduate Researcher - Molecular Neuroscience

September 2014 - March 2016 (1 year 7 months)

Jerusalem, Israel

Conducted research on microRNA regulation and gene expression using bioinformatics and data analysis tools, studying molecular mechanisms of neural communication.

Education

Weizmann Institute of Science

Master of Science - MS, Systems Neuroscience · (October 2017 - November 2019)

The Hebrew University

Bachelor’s Degree, Physiological Psychology/Psychobiology · (2013 - 2016)

University of Melbourne

Exchange program, Physiological Psychology/Psychobiology · (2016 - 2016)